Single Sine Excitation of a Nonlinear Activation Unit: Arbitrary Gain Signals

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Setup

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clc; clear; close all;
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Problem

A nonlinear activation unit with input gain γ is defined:

$$\varphi(v) = \tanh(\gamma\, v)$$

For input signal $v = \sin(\omega t + \phi\ )$ we wish to know the Fourier Series of the output signal $y = \varphi(v)$

That is we want to know the Fourier Series of:

$$y(t) = \tanh(\gamma \sin(\omega t + \phi\ ))$$

When the input gain is restricted to $|\gamma| < \frac{\pi}{2}$, the following expression can be derived via straightforward Taylor series expansion, the power reduction formula for trigonometric polynomials, and series manipulations:

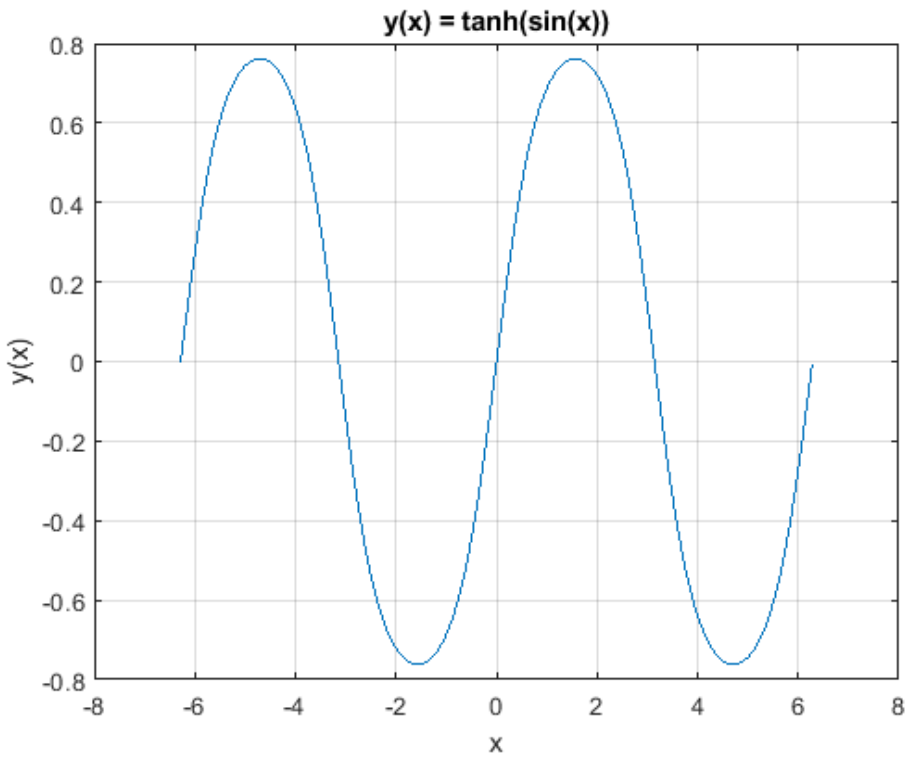
$$y(x) = \sum_{n=1}^{\infty} \sum_{m=n}^{\infty} \frac{(-1)^{2m-n-1} 4(2^{2m}-1)\gamma^{\,2m-1} C_{m-n}^{2m-1} B_{2m}}{(2m)!} \sin((2n-1)x)$$

Now we desire an expression for the Fourier Series of $y(t)$ for all $\gamma > 0$.

Classic Simplifications using Symmetry

First note that the function $y(x) = \tanh(\gamma \sin(x))$ is odd:

```
x = -2*pi:0.01:2*pi; y = tanh(sin(x));
figure; plot(x,y); xlabel('x'); ylabel('y(x)'); grid on; title('y(x) = tanh(sin(x))');
```



Now note the definition of the Fourier Series for a 2π periodic function:

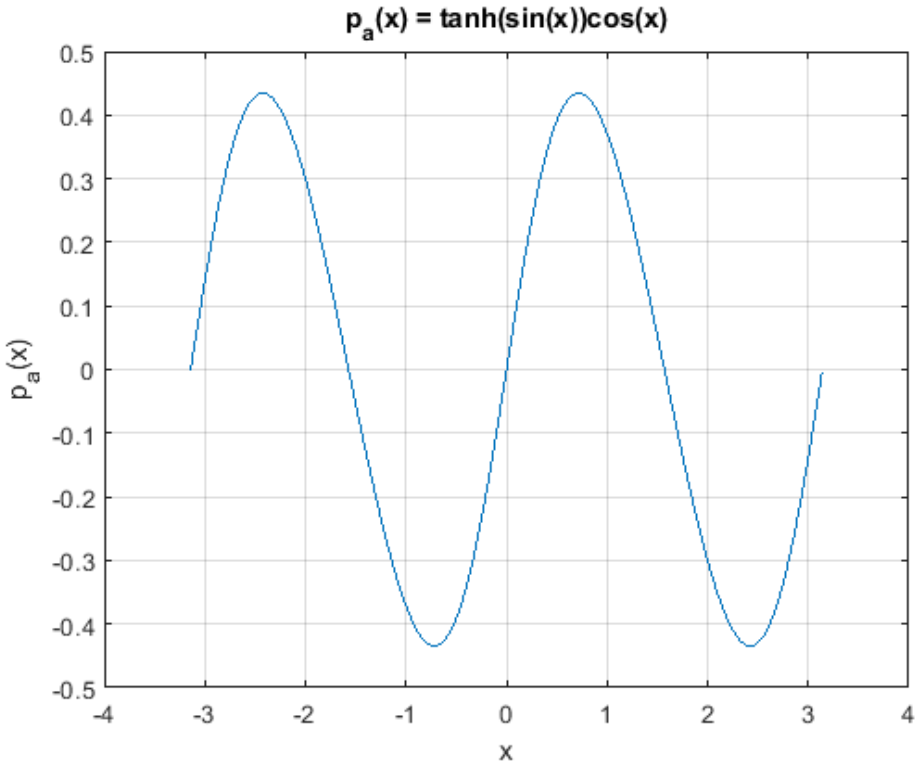
$$y(x) = \frac{1}{2}a_0 + \sum_{n=1}^{\infty} a_n \cos(n\,x) + \sum_{n=1}^{\infty} b_n \sin(n\,x)$$

Where:

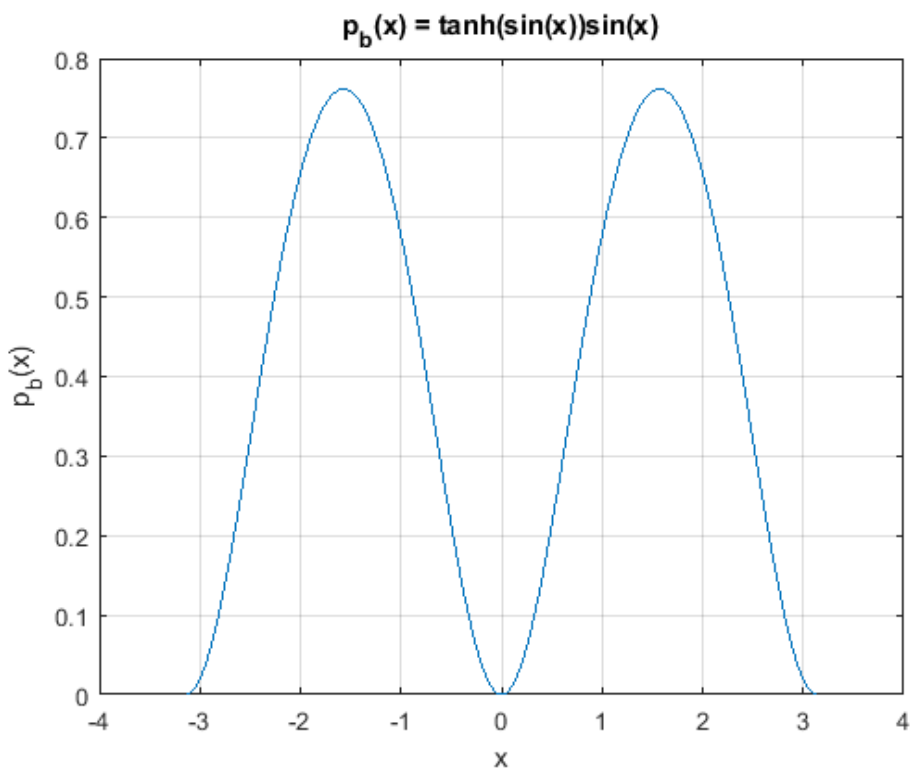
$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} y(x)\mathrm{dx},\;\; a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} y(x)\cos(\mathrm{nx})\mathrm{dx},\;\; b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} y(x)\sin(n\,x)\mathrm{dx}$$

Now note $p_a(x) = \tanh(\sin(x))\cos(n\,x)$ is a product of an odd and an even function, and is therefore odd. Likewise, $p_b(x) = \tanh(\sin(x))\sin(n\,x)$ is a product of an odd function with another odd function, and is therefore even.

```
x = -pi:0.01:pi; pa = tanh(sin(x)).*cos(x); pb = tanh(sin(x)).*sin(x);
figure; plot(x,pa); xlabel('x'); ylabel('p_a(x)'); grid on; title('p_a(x) = tanh(sin(x))cos(x)');
```




```
figure; plot(x,pb); xlabel('x'); ylabel('p_b(x)'); grid on; title('p_b(x) = tanh(sin(x))sin(x)');
```



The above plots make the following classic Fourier Series identities for odd $y(x)$ obvious:

$$a_0 = \frac{1}{\pi}\int_{-\pi}^{\pi} y(x)\mathrm{dx} = 0$$

$$a_n = \frac{1}{\pi}\int_{-\pi}^{\pi} y(x)\cos(n\,x)\mathrm{dx} = 0$$

$$b_n = \frac{1}{\pi}\int_{-\pi}^{\pi} y(x)\sin(n\,x)\mathrm{dx} = \frac{2}{\pi}\int_{0}^{\pi} y(x)\sin(n\,x)\mathrm{dx}$$

Thus we can write an initial expression for the Fourier Series of $y(x)$:

$$y(x) = \tanh(\gamma\,\sin(x)) = \frac{2}{\pi}\sum_{n=1}^{\infty}\left\{\int_0^{\pi}\tanh(\gamma\,\sin(x))\sin(n\,x)\mathrm{dx}\right\}\sin(n\,x)$$

Additionally, because the hyperbolic tangent is an odd nonlinear function, we know that all the even components of the Fourier Series will be zero:

$$y(x) = \tanh(\gamma\,\sin(x)) = \frac{2}{\pi}\sum_{n=1}^{\infty}\left\{\int_0^{\pi}\tanh(\gamma\,\sin(x))\sin((2n-1)\,x)\mathrm{dx}\right\}\sin((2n-1)\,x)$$

%%%

$$\int_0^{\pi} \tanh(\gamma \sin(x))\sin((2n-1)\,x)\mathrm{dx} = \int_0^{\pi} I_1(x)\mathrm{dx}$$

We can start with rule 1.232.1 on page 27 (page 76 in the PDF version) of Gradsteyn and Rhyzik:

$$\tanh(v) = 1 + 2\sum_{m=1}^{\infty} (-1)^m e^{-2\,m\,v} \quad \forall\, v > 0$$

Giving:

$$I_1(x) = \left(1 + 2\sum_{m=1}^{\infty} (-1)^m e^{-2\,m\,\gamma\,\sin(x)}\right)\sin((2n-1)\,x)$$

Therefore we have:

$$\int_0^{\pi} \tanh(\gamma \sin(x))\sin((2n-1)\,x)\mathrm{dx} = \int_0^{\pi} \sin((2n-1)\,x)\mathrm{dx} + \int_0^{\pi} \left\{2\sin((2n-1)\,x)\sum_{m=1}^{\infty} (-1)^m e^{-2\,m\,\gamma\,\sin(x)}\right\}\mathrm{dx}$$

Evaluating the first integral and rearranging the second:

$$\int_0^{\pi} \tanh(\gamma \sin(x))\sin((2n-1)\,x)\mathrm{dx} = \frac{-\cos(\pi(2n-1))+1}{2n-1} + \sum_{m=1}^{\infty}\left\{2(-1)^m \int_0^{\pi} \sin((2n-1)x)e^{-2\,m\,\gamma\,\sin(x)}\,\mathrm{dx}\right\}$$

Simplifying the cosine expression:

$$\frac{-\cos(\pi(2n-1))+1}{2n-1} = \frac{2}{2n-1}$$

$$\int_0^{\pi} \tanh(\gamma \sin(x))\sin((2n-1)\,x)\mathrm{dx} = \frac{2}{2n-1} + \sum_{m=1}^{\infty}\left\{2(-1)^m \int_0^{\pi} \sin((2n-1)\,x)e^{-2\,m\,\gamma\,\sin(x)}\,\mathrm{dx}\right\}$$

Substituting the new expression for $\int_0^{\pi} \tanh(\gamma \sin(x))\sin((2n-1)\,x)\mathrm{dx}$ into Theorem 1 gives Theorem 2:

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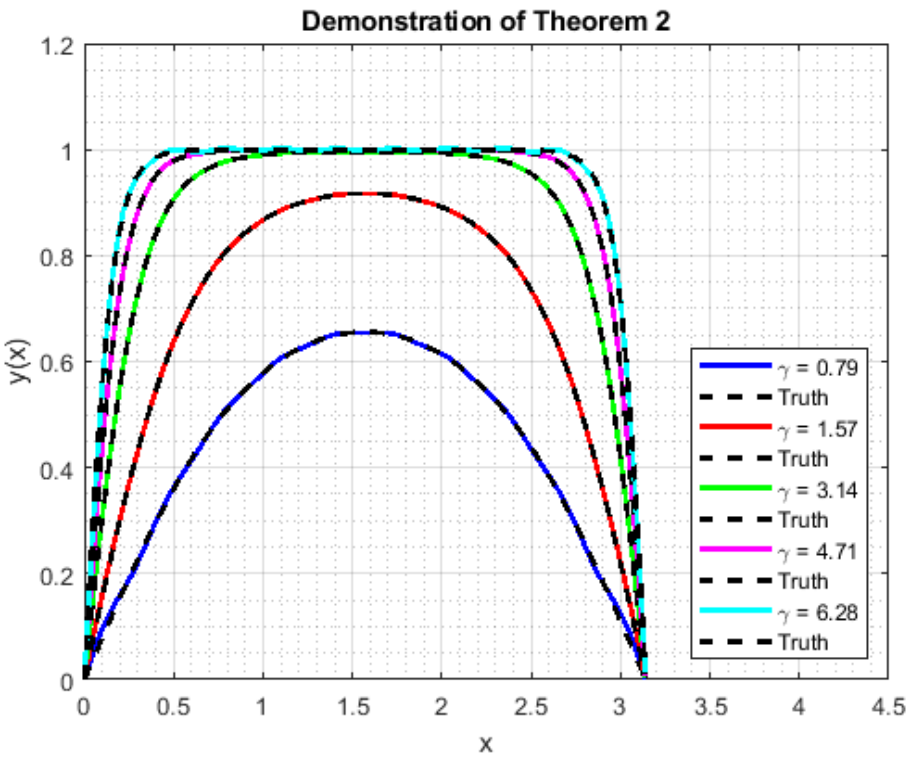

Evaluating the Reduced Integral

Now we have reduced evaluating the above expression to evaluating the reduced integral:

$$\int_0^{\pi} \sin((2n-1)\,x) e^{-2\,m\,\gamma\,\sin(x)}\,\mathrm{dx} = \int_0^{\pi} I_2(x)\,\mathrm{dx}$$

We start by taking the Taylor Series for $e^{-2\,m\,\gamma\,\sin(x)}$ using:

$$e^{-2\,m\,v} = \sum_{l=0}^{\infty} \frac{(-2\,m\,v)^l}{l!}$$

Then the integrand is:

$$I_2(x) = \sin((2n-1)\,x) \sum_{l=0}^{\infty} \frac{(-2\,m\,\gamma\,\sin(x))^l}{l!}$$

Giving:

$$\int_0^{\pi} \sin((2n-1)\,x) e^{-2\,m\,\gamma\,\sin(x)}\,\mathrm{dx} = \int_0^{\pi} \sin((2n-1)\,x) \sum_{l=0}^{\infty} \frac{(-2\,m\,\gamma\,\sin(x))^l}{l!}\mathrm{dx}$$

Moving the summation symbol outside the integral:

$$\int_0^{\pi} \sin((2n-1)\,x) e^{-2\,m\,\gamma\,\sin(x)}\,\mathrm{dx} = \sum_{l=0}^{\infty} \int_0^{\pi} \sin((2n-1)\,x) \frac{(-2\,m\,\gamma\,\sin(x))^l}{l!}\,\mathrm{dx}$$

Distributing the exponent:

$$\int_0^{\pi} \sin((2n-1)\,x) e^{-2\,m\,\gamma\,\sin(x)}\,\mathrm{dx} = \sum_{l=0}^{\infty} \int_0^{\pi} \sin((2n-1)\,x) \frac{(-2\,m\,\gamma)^l \sin^l(x)}{l!}\,\mathrm{dx}$$

Moving all terms that are constant with respect to x outside the integral:

$$\int_0^{\pi} \sin((2n-1)\,x) e^{-2\,m\,\gamma\,\sin(x)}\,\mathrm{dx} = \sum_{l=0}^{\infty} \frac{(-2\,m\,\gamma)^l}{l!} \int_0^{\pi} \sin((2n-1)x) \sin^l(x)\mathrm{dx}$$

Substituting back into Theorem 2:

$$\tanh(\gamma\sin(x)) = \frac{2}{\pi}\sum_{n=1}^{\infty}\left\{\frac{2}{2n-1}+\sum_{m=1}^{\infty}\left\{2(-1)^m\left\{\sum_{l=0}^{\infty}\frac{(-2\,m\,\gamma)^l}{l!}\int_0^{\pi}\sin((2n-1)\,x)\sin^l(x)\mathrm{dx}\right\}\right\}\right\}\sin((2n-1)\,x)$$

Rearranging the summations:

$$\tanh(\gamma\sin(x)) = \frac{2}{\pi}\sum_{n=1}^{\infty}\left\{\frac{2}{2n-1}+\sum_{m=1}^{\infty}\sum_{l=0}^{\infty}\frac{2(-1)^m(-2\,m\,\gamma)^l}{l!}\int_0^{\pi}\sin((2n-1)\,x)\sin^l(x)\mathrm{dx}\right\}\sin((2n-1)\,x)$$

Cleaning up the indicies inside the summation over l and m:

$$\tanh(\gamma\sin(x)) = \frac{2}{\pi}\sum_{n=1}^{\infty}\left\{\frac{2}{2n-1}+\sum_{m=1}^{\infty}\sum_{l=0}^{\infty}\frac{(-1)^{m+l}2^{l+1}m^l}{l!}\gamma^l\int_0^{\pi}\sin((2n-1)\,x)\sin^l(x)\mathrm{dx}\right\}\sin((2n-1)\,x)$$

%%%

Algorithms

```
% @breif Function for testing theorem 1 using numerical integration
% @details Numerically computes the Fourier Series of tanh(g*sin(x)) for
% several values of g.
% @param[in] n_max - Truncation of the Fourier Series
```

```
function TestTheorem1(n_max)
    x = 0:0.01:pi;
    gamma_list = [pi/4 pi/2 pi   3*pi/2 2*pi];
    colors     = ['b', 'r', 'g', 'm',   'c'];
    legend_strs = {};
    figure;
    for gamma_idx = 1:length(gamma_list)
        sum = 0;
        for n = 1:n_max
            integrand = tanh(gamma_list(gamma_idx)*sin(x)).*sin((2*n-1)*x);
            integral = trapz(x, integrand);
            sum = sum + integral*sin((2*n-1)*x);
        end
        y_approx = 2/pi*sum;
        y_truth = tanh(gamma_list(gamma_idx)*sin(x));
        plot(x, y_approx, '-', 'color', colors(gamma_idx), 'linewidth', 2);
        hold on;
        plot(x, y_truth, 'k--', 'linewidth', 2);
        legend_str = sprintf('\\gamma = %0.2f', gamma_list(gamma_idx));
        legend_strs{end+1} = legend_str;
        legend_strs{end+1} = 'Truth';
    end
    legend(legend_strs, 'location', 'southeast');
    grid on;
    grid minor;
    xlabel('x');
    ylabel('y(x)');
    title('Demonstration of Theorem 1');
    xlim([0 4.5]);
end

% @breif Function for testing theorem 2 using numerical integration
% @details Numerically computes the Fourier Series of tanh(g*sin(x)) for
% several values of g.
% @param[in] n_max - Truncation of the Fourier Series
% @param[in] m_max - Truncation of hyperbolic tangent's Taylor Series
function TestTheorem2(n_max, m_max)
    x = 0:0.01:pi;
    gamma_list = [pi/4 pi/2 pi   3*pi/2 2*pi];
    colors     = ['b', 'r', 'g', 'm',   'c'];
    legend_strs = {};
    figure;
    for gamma_idx = 1:length(gamma_list)
        outer_sum = 0;
        for n = 1:n_max
            inner_sum = 0;
            for m = 1:m_max
                integrand = sin((2*n-1)*x).*exp(-2*m*gamma_list(gamma_idx)*sin(x));
                integral = trapz(x, integrand);
                inner_sum = inner_sum + 2*(-1)^m*integral;
            end
```

```
            outer_sum = outer_sum + (  2/(2*n-1) + inner_sum  )*sin((2*n-1)*x);
        end
        y_approx = 2/pi*outer_sum;
        y_truth = tanh(gamma_list(gamma_idx)*sin(x));
        plot(x, y_approx, '-', 'color', colors(gamma_idx), 'linewidth', 2);
        hold on;
        plot(x, y_truth, 'k--', 'linewidth', 2);
        legend_str = sprintf('\\gamma = %0.2f', gamma_list(gamma_idx));
        legend_strs{end+1} = legend_str;
        legend_strs{end+1} = 'Truth';
    end
    legend(legend_strs, 'location', 'southeast');
    grid on;
    grid minor;
    xlabel('x');
    ylabel('y(x)');
    title('Demonstration of Theorem 2');
    xlim([0 4.5]);
end

% @breif Function for testing theorem 3 using numerical integration
% @details Numerically computes the Fourier Series of tanh(g*sin(x)) for
% several values of g.
% @param[in] n_max - Truncation of the Fourier Series
% @param[in] m_max - Truncation of hyperbolic tangent's Taylor Series
% @param[in] l_max - Truncation of exponential function's Taylor Series
function TestTheorem3(n_max, m_max, l_max)
    x = 0:0.01:pi;
    gamma_list = [pi/4 pi/2 pi 3*pi/2];
    colors     = ['b'  'r' 'g' 'm'];
    legend_strs = {};
    figure;
    for gamma_idx = 1:length(gamma_list)
        n_sum = 0;
        for n = 1:n_max
            m_sum = 0;
            for m = 1:m_max
                l_sum = 0;
                for l = 0:l_max
                    gamma_power = gamma_list(gamma_idx)^(l);
                    q = (-1)^(m+l)*(2)^(l+1)*m^l/factorial(l);
                    integrand = sin((2*n-1)*x).*(sin(x).^(l));
                    integral = trapz(x, integrand);
                    l_sum = l_sum + q*integral*gamma_power;
                end
                m_sum = m_sum + l_sum;
            end
            n_sum = n_sum + (2/(2*n-1) + m_sum)*sin((2*n-1)*x);
        end
        y_approx = 2/pi*n_sum;
        y_truth = tanh(gamma_list(gamma_idx)*sin(x));
        plot(x, y_approx, '-', 'color', colors(gamma_idx), 'linewidth', 2);
        hold on;
        plot(x, y_truth, 'k--', 'linewidth', 2);
        legend_str = sprintf('\\gamma = %0.2f', gamma_list(gamma_idx));
        legend_strs{end+1} = legend_str;
        legend_strs{end+1} = 'Truth';
    end
```

```
    legend(legend_strs, 'location', 'southeast');
    grid on;
    grid minor;
    xlabel('x');
    ylabel('y(x)');
    title('Demonstration of Theorem 3');
    xlim([0 4.5]);
end
```